

ICT 332

Generative AI & Large Language Models

LECTURE STUDY NOTES

Weeks 8 – 14

Prepared as comprehensive study notes covering all topics, concepts, key terms, and learning outcomes from Week 8 through Week 14 of the ICT332 module.

Topics Covered:

- Week 8 — Mid-Semester Assessment & Practical Competency Evaluation
- Week 9 — AI APIs, Application Development & Building AI-Powered Systems
- Week 10 — Retrieval-Augmented Generation (RAG), Embeddings & Knowledge-Grounded AI
- Week 11 — Multimodal AI Systems, Image Generation, Speech AI & Vision-Language Models
- Week 12 — Ethics, Governance, Regulation & Responsible AI
- Week 13 — AI Innovation for Zambia & Africa, Entrepreneurship & Local AI Solutions
- Week 14 — Final Project Demonstrations, Viva Presentations & Course Revision

WEEK 8 — MID-SEMESTER ASSESSMENT & PRACTICAL COMPETENCY EVALUATION

Theme: Demonstrating Practical Understanding, Critical Thinking, and Responsible AI Usage

1. Introduction — What Is This Week About?

Week 8 is your mid-semester checkpoint. Instead of a traditional exam that tests memorisation, this week evaluates what you can **do**, **explain**, **evaluate**, and **defend**. Think of it as a performance review — you are being assessed on genuine competence, not just knowing facts by heart.

Key message: You are assessed not only on 'what you know' but also on 'what you can DO, EXPLAIN, EVALUATE, and DEFEND.'

2. The Four Principles of This Assessment

Principle	What It Means
Authenticity	You must show genuine understanding — not just polished-looking AI-generated answers.
Practicality	Tasks mirror real AI workflows that professionals use every day.
Accountability	You must be able to explain and defend every piece of work you submit.
Competence	The focus is on applied skills — can you actually use AI responsibly?

3. Assessment Structure

The mid-semester assessment has THREE components:

Component	Weighting	Focus
Part A — Theory Assessment	40%	Conceptual understanding, critical reasoning, ethical awareness
Part B — Practical Laboratory Assessment	40%	Hands-on competence, prompt engineering, workflow design
Part C — Oral Demonstration / Viva	20%	Genuine understanding, ability to explain and defend work

4. Part A — Theory Assessment (40%)

The theory component tests your ability to think, analyse, and explain AI concepts. Questions are not simple recall — they require you to reason, compare, and evaluate scenarios. Duration is typically 1.5–2 hours.

Types of Questions You Will Face:

Short Answer: Define key terms — e.g., 'What is a hallucination in AI?', 'Define prompt engineering.'

Analytical: Given a real scenario (e.g., a chatbot giving wrong medical advice), discuss causes, risks, and solutions.

Comparative: Compare two concepts — e.g., zero-shot vs few-shot prompting, open-source vs proprietary AI.

Critical Reflection: Argue both sides of an issue — e.g., 'Should students use AI in education?'

Applied Scenario: Design a practical solution — e.g., an AI advisory system for Zambian farmers.

5. Part B — Practical Laboratory Assessment (40%)

In the lab you complete real tasks using AI tools. You are evaluated on how well you design prompts, catch errors, iterate, and think like an AI developer.

- Task 1: Prompt Engineering Challenge — Take a vague prompt and improve it for a specific audience, adding structure and clarity.
- Task 2: Hallucination Detection — Analyse AI-generated text that contains errors or fabricated citations. Identify them and suggest fixes.
- Task 3: Prompt Iteration Exercise — Start with an initial prompt, evaluate the output, refine the prompt, and show the improvement.
- Task 4: AI Workflow Design — Map out a full workflow (e.g., research summariser, tutoring bot) including verification and human oversight steps.
- Task 5: Bias Analysis — Examine AI outputs for cultural bias, gender assumptions, or misinformation.
- Task 6: Structured Output Generation — Use AI to produce tables, JSON data, reports, or formatted documents.

Practical Rubric at a Glance:

Criterion	Marks
Prompt Design Quality	20
Critical Evaluation	20
Practical Competence	20
Workflow Design	15
Ethical Awareness	10
Reflection and Explanation	15

6. Part C — Oral Demonstration / Viva (20%)

The viva is a 5–10 minute oral examination where the lecturer asks you to explain your work. This is designed to confirm that you genuinely understand what you submitted — students who relied entirely on AI-generated work will struggle here. You must be able to explain: why you wrote the prompt that way, what caused a hallucination, how your workflow works, and what ethical risks you identified.

Important: Students who cannot explain their prompts, workflows, or outputs may lose marks even if their written submission looks polished. Be honest and prepared.

WEEK 9 — AI APIs, APPLICATION DEVELOPMENT & BUILDING AI-POWERED SYSTEMS

Theme: Moving from AI Users to AI Builders

1. The Big Idea — From User to Builder

Up until now you have been *using* AI systems. This week you cross a very important line — you start *building* AI-powered applications. Modern AI systems are rarely trained from scratch. Instead, developers access powerful pre-trained models through **APIs** (Application Programming Interfaces) and build products on top of them.

Core concept: Modern AI systems are most often accessed through APIs rather than trained from scratch. This means a student or small startup can build powerful AI applications without needing a supercomputer.

2. What Is an API?

An **API (Application Programming Interface)** is a set of rules that allows one software system to talk to another. Think of it like a waiter in a restaurant: you (the application) give the waiter (the API) your order; the waiter takes it to the kitchen (the AI model); and the kitchen sends back your food (the response). You never go into the kitchen directly.

When working with AI APIs you typically send an HTTP request containing your prompt and settings, and receive a structured response in **JSON format**. You authenticate using an **API key** — a private credential that identifies your account and controls your access.

3. Key AI APIs You Should Know

OpenAI API: Provides access to GPT-4, GPT-3.5, DALL-E, and Whisper. Industry standard for many commercial AI apps.

Hugging Face APIs: Open-source hub hosting thousands of models for text, images, and audio. Great for experimentation.

Google Gemini API: Google's multimodal AI API — handles text, images, audio, and code.

Anthropic (Claude) API: Access to Claude models, known for safety and helpfulness.

Local AI (Ollama): Run AI models entirely on your own computer — no internet required. Great for privacy.

4. AI Application Architecture

When you build an AI-powered application, it typically has these layers working together:

Frontend (User Interface): What the user sees and interacts with — a web page, mobile app, or chatbot window.

Backend (Server Logic): Processes user requests, calls the AI API, handles business logic, and manages data.

AI API Layer: The external AI model that generates responses based on prompts.

Database: Stores user data, conversation history, or retrieved documents.

Verification/Human Oversight: Checks AI outputs for accuracy, safety, and compliance before showing users.

5. Building AI Chatbots

A chatbot is one of the most common AI applications. Key components include:

- Conversation history management — the AI needs context from earlier in the conversation to give coherent replies.
- Prompt pipelines — a series of instructions that guide the AI's behaviour and persona.
- Error handling — graceful responses when the API fails, times out, or produces unhelpful outputs.
- User experience design — clear, simple interfaces so users can interact naturally.
- Fallback responses — pre-written replies for when the AI cannot answer confidently.

6. AI-Assisted Workflow Automation

Beyond chatbots, AI APIs can automate entire workflows: summarising documents, generating reports, classifying customer emails, or drafting responses. These systems chain multiple AI calls together with logic in between — a concept sometimes called a **prompt pipeline** or **AI workflow**.

7. Deployment Concepts

Cloud Deployment: Run your AI app on services like AWS, Google Cloud, or Azure. Scalable but costs money.

Local Deployment: Run on your own machine. Free and private but limited by hardware.

Edge AI: Deploy lightweight AI models on devices (phones, sensors). Useful where internet is limited — very relevant for Africa.

8. Limitations to Remember

Common pitfalls of AI applications: Latency (slow responses), Hallucinations (still present), Cost (API calls cost money), Rate Limits (APIs restrict how many requests you can make), Reliability (models can behave unpredictably with certain prompts).

9. Security & Privacy

- Never expose your API key in public code or repositories — treat it like a password.
- Collect only the user data you actually need.
- Be transparent with users when they are interacting with AI, not a human.
- Comply with data protection laws when handling personal information.

KEY TERM	MEANING
API	Application Programming Interface — rules for software-to-software communication
Endpoint	A specific URL where an API service is available
JSON	JavaScript Object Notation — a structured data format for API responses
API Key	A private credential authenticating your access to an API

Chatbot	A conversational AI application that interacts with users in natural language
Backend	Server-side application logic, hidden from the user
Frontend	The user-facing interface of an application
Workflow Automation	Using AI to automatically execute sequences of tasks
Rate Limit	A cap on how many API requests you can make per unit of time
LangChain	A popular Python framework for building AI-powered applications

WEEK 10 — RETRIEVAL-AUGMENTED GENERATION (RAG), EMBEDDINGS & KNOWLEDGE-GROUNDED AI

Theme: Improving AI Reliability Using External Knowledge and Retrieval Systems

1. The Problem — Why LLMs Need Help

Large Language Models (LLMs) have a fundamental limitation: their knowledge is frozen at the point they were trained. If you ask an LLM about something that happened after its training cutoff, or about a specific internal document your organisation has, it simply does not know — and worse, it may hallucinate a plausible-sounding but wrong answer.

The three core problems are: **Knowledge Cutoffs** (information from before training only), **Limited Memory** (LLMs cannot store your specific documents), and **Hallucinations** (confident wrong answers).

2. The Solution — Retrieval-Augmented Generation (RAG)

RAG = Retrieval + AI Generation. Instead of relying only on what the model memorised during training, a RAG system RETRIEVES relevant information from an external source first, then gives that information to the LLM as context so it can generate a grounded, accurate answer.

Think of it like this: imagine you are sitting an open-book exam. Without RAG, the LLM must answer from memory alone (and may guess wrong). With RAG, the LLM can look up the right pages of the textbook first, then write its answer based on real information.

3. How RAG Works — Step by Step

Step 1 — Document Ingestion: Your documents (PDFs, web pages, notes) are loaded into the system.

Step 2 — Chunking: Documents are split into smaller pieces (chunks) — e.g., paragraphs or sections. This is important because context windows have size limits.

Step 3 — Embedding Generation: Each chunk is converted into a numerical vector (an embedding) that captures its meaning.

Step 4 — Vector Storage: These embeddings are stored in a vector database (e.g., ChromaDB, FAISS).

Step 5 — User Query: When a user asks a question, their question is also converted into an embedding.

Step 6 — Similarity Search: The system finds chunks whose embeddings are most similar (semantically) to the query.

Step 7 — Context Injection: The relevant chunks are inserted into the prompt as context for the LLM.

Step 8 — Grounded Response: The LLM generates an answer based on the retrieved evidence, not just its memory.

Step 9 — Human Verification: The output should still be checked, as retrieval errors can still occur.

4. Embeddings — The Secret Sauce

An **embedding** is a list of numbers (a vector) that represents the meaning of a word, sentence, or document. The key insight is that semantically similar pieces of text will have embeddings that are mathematically close together in vector space.

For example, the sentences 'How to grow maize' and 'Techniques for cultivating corn' are about the same topic even though they use different words. Their embeddings will be close together, so a semantic search will correctly identify them as related.

Important: Embeddings represent statistical relationships in language data — they do NOT 'understand' meaning the way a human does. They are a very clever mathematical approximation of meaning.

5. Keyword Search vs Semantic Search

Feature	Keyword Search	Semantic Search
How it works	Matches exact words	Matches meaning/concept
Example query	'maize farming'	'How can I improve crop yield?'
Handles synonyms?	No	Yes
Handles context?	Limited	Yes
Best for	Exact document lookup	Conceptual, conversational queries

6. Document Chunking — Why It Matters

When splitting documents into chunks, you need to balance two things: chunks that are too small may lack enough context to be useful; chunks that are too large may exceed the LLM's context window or include irrelevant information. There is also the concept of **overlap** — allowing adjacent chunks to share a few sentences to ensure no important context is cut off at boundaries.

7. Limitations of RAG

- RAG does NOT eliminate hallucinations completely — if the retrieved chunk is wrong or irrelevant, the LLM may still generate incorrect answers.
- Poor chunking can break important context and reduce retrieval quality.
- Semantic search still has limits — ambiguous queries may retrieve irrelevant chunks.
- The quality of the knowledge base matters enormously — garbage in, garbage out.
- Human verification remains essential even in RAG systems.

KEY TERM	MEANING
RAG	Retrieval-Augmented Generation — combining retrieval with AI generation
Embedding	A numerical vector representation of text meaning
Vector Database	A specialised database for storing and searching embeddings (e.g., ChromaDB, FAISS)
Semantic Search	Search based on meaning rather than exact keyword matching
Chunking	Splitting documents into smaller sections for retrieval
Grounding	Linking AI outputs to verified, retrieved knowledge
Knowledge Base	A collection of documents the RAG system can retrieve from

Similarity Search	Finding embeddings that are mathematically close to a query embedding
Context Window	The maximum amount of text an LLM can process at once
LlamaIndex	A Python framework for building RAG applications

WEEK 11 — MULTIMODAL AI SYSTEMS, IMAGE GENERATION, SPEECH AI & VISION-LANGUAGE MODELS

Theme: Understanding AI Systems That Combine Text, Images, Audio, Video, and Vision

1. Beyond Text — What Is Multimodal AI?

Most of what you have studied so far has involved text-in, text-out AI systems. **Multimodal AI** breaks this limitation — these systems can process and generate multiple types of information simultaneously, including text, images, audio, speech, and video. This mirrors how humans naturally communicate: we do not only use words — we use pictures, gestures, sounds, and more.

Examples of multimodal AI systems you may already know: GPT-4V (reads images AND text), DALL-E (generates images from text descriptions), Whisper (transcribes speech to text), Google Gemini (understands text, images, audio, and video simultaneously).

2. Text-to-Image Generation

These systems generate images from text descriptions (prompts). The dominant technology behind them is called **diffusion models**. Examples include DALL-E, Stable Diffusion, and Midjourney.

How diffusion works (simplified): The model starts with random noise (imagine static on a TV screen). It has learned during training how to gradually 'de-noise' an image — removing the static step by step — guided by your text prompt, until a coherent image emerges.

Prompting for images: The quality of image generation depends heavily on how you describe what you want. Good image prompts include: subject, style, lighting, composition, and specific details. Example: instead of 'a farmer', try 'a Zambian farmer in a green maize field at sunrise, photorealistic, warm golden lighting.'

3. Vision-Language Models (VLMs)

A Vision-Language Model is an AI that can both see images AND understand text — and reason about both together. Tasks that VLMs can perform include:

- Image captioning — describing what is in an image in words.
- Visual question answering (VQA) — answering questions about the contents of an image.
- Object detection — identifying and locating specific objects in images.
- Document understanding — reading text from photographs of documents (e.g., receipts, signs).
- Medical imaging analysis — helping doctors identify patterns in X-rays or scans.

4. Speech AI Systems

Speech Recognition (STT — Speech-to-Text)

Converts spoken words into written text. Example: Whisper by OpenAI. Key challenges: accents, background noise, rare languages, local dialects (like Bemba or Nyanja).

Text-to-Speech (TTS)

Converts written text into spoken audio. Modern systems sound remarkably natural. Examples: ElevenLabs, Google TTS, Microsoft Azure Speech.

Voice Assistants

Combine both: they listen to you (STT), understand your intent (LLM), and speak back (TTS). Examples: Siri, Alexa, Google Assistant.

5. Optical Character Recognition (OCR)

OCR is the technology that extracts text from images. If you photograph a handwritten letter, a printed document, or a street sign, OCR converts what it sees into editable, searchable text. This is extremely valuable for digitising records in contexts where physical documents are still common. Limitations include difficulty with poor handwriting, unusual fonts, and low-quality images.

6. Deepfakes and Synthetic Media — A Serious Ethical Concern

The same technology that creates impressive multimodal AI can also be weaponised to create **deepfakes** — highly realistic but completely fabricated videos, audio recordings, or images of real people doing or saying things they never actually did.

Critical thinking point: Deepfakes are not just entertaining novelties. They can be used to spread political misinformation, damage reputations, commit fraud, manipulate elections, or harm individuals. The ability to critically evaluate digital media for authenticity is now an essential life skill.

How to spot deepfakes: Look for unnatural eye blinking, inconsistent lighting on faces, blurry or distorted edges around hair, audio that does not quite match lip movements, or background elements that flicker unnaturally.

7. Ethical Challenges in Multimodal AI

- Copyright and intellectual property — who owns an AI-generated image? Can AI legally reproduce an artist's style?
- Synthetic media labelling — should AI-generated content be legally required to be labelled as such?
- Deepfake misinformation — fake media used in politics, journalism, and social media.
- Biases in image generation — AI image models often reflect and amplify stereotypes from their training data.
- Privacy violations — generating realistic images of real people without consent.

KEY TERM	MEANING
Multimodal AI	AI systems that process and generate multiple types of data (text, image, audio, video)
Diffusion Model	AI model architecture used for image generation — progressively removes noise
Vision-Language Model (VLM)	AI combining visual understanding with language processing
OCR	Optical Character Recognition — extracting text from images
Text-to-Speech (TTS)	Generating spoken audio from written text
Speech Recognition (STT)	Converting spoken audio into written text

Deepfake	AI-generated synthetic media that convincingly impersonates real people
Synthetic Media	Any AI-generated digital content — images, audio, video, or text
Image Captioning	Automatically generating a text description of an image
Visual QA	Answering questions about the contents of an image

WEEK 12 — ETHICS, GOVERNANCE, REGULATION & RESPONSIBLE AI

Theme: Ensuring AI Systems Are Fair, Safe, Transparent, and Beneficial to Society

1. Why Ethics in AI Matters

Throughout this course you have seen that AI systems can hallucinate, generate harmful content, amplify bias, and manipulate information. Now we ask the deeper question: **what are our responsibilities when building, deploying, and using these systems?**

Core insight: AI is not only a technical issue — it is also a human, political, legal, and societal issue. Building powerful AI without ethical oversight can create serious social, economic, and human harms.

2. The Core Ethical Principles

Fairness: AI systems should treat all people equitably. They should not discriminate based on race, gender, age, disability, or geography.

Transparency: Users and affected parties should be able to understand how AI systems make decisions. Black-box systems that cannot be explained are problematic.

Accountability: When an AI system causes harm, someone must be responsible. Developers, companies, and governments all share accountability.

Privacy: AI systems must respect people's personal data. Collection, storage, and use of data must be consensual and lawful.

Safety: AI systems must not cause physical, psychological, or societal harm. Safety testing is essential before deployment.

Human Oversight: Humans must remain in control of consequential AI decisions — especially in healthcare, law, finance, and governance.

Reliability: AI systems should perform consistently and accurately, especially in high-stakes environments.

3. Algorithmic Bias — A Detailed Look

Algorithmic bias occurs when an AI system produces systematically unfair outcomes for certain groups of people. This typically happens because:

- Training data that over-represents some groups and under-represents others (e.g., most medical datasets historically focused on Western populations).
- Datasets that reflect historical human discrimination — the AI learns and perpetuates those patterns.
- Poor diversity among the teams building the AI — different perspectives catch different problems.
- Objectives (what the model is optimised for) that do not account for fairness.

Real-world example: An AI hiring tool trained on historical hiring data at a company that historically favoured male candidates will learn to rank male candidates higher — systematically disadvantaging equally qualified women.

4. Explainable AI (XAI)

The challenge with many modern AI systems is that they are '**black boxes**' — even their designers cannot fully explain why they produced a specific output. This is a serious problem in high-stakes decisions like medical diagnosis, loan approvals, or criminal justice risk assessments.

Explainable AI (XAI) is a field of research focused on making AI decisions interpretable to humans. The goal is that when an AI rejects your bank loan or diagnoses a medical condition, you — or a regulator — can understand why and challenge the decision if it is wrong.

5. AI Governance and Regulation

Governments and international organisations are developing frameworks to regulate AI. The general approach is **risk-based governance**: the higher the risk of harm, the stricter the rules.

- European Union AI Act — one of the world's most comprehensive AI regulation frameworks, categorising AI applications by risk level.
- UNESCO AI Ethics Recommendation — a global framework for responsible AI development.
- Voluntary commitments by major AI companies to safety testing and transparency.

Key governance questions: Who is responsible when an AI system causes harm? Should AI decisions be challengeable in court? Should powerful AI systems require government approval before deployment?

6. AI and Society — Broader Impacts

Employment: AI automation may displace workers in certain industries. However, AI also creates new jobs and opportunities — the net effect depends on how societies adapt.

Education: AI tutors, grading systems, and content generators are transforming education. Questions of academic integrity, critical thinking development, and equitable access arise.

Democracy & Information: AI-generated misinformation and deepfakes can undermine democratic processes and public trust in institutions.

Digital Inequality: If AI systems are only accessible to wealthy, connected populations, they may widen existing inequalities.

Surveillance: AI facial recognition and behavioural monitoring raise serious privacy and civil liberties concerns.

KEY TERM	MEANING
AI Ethics	Moral principles and values guiding responsible AI development and use
Algorithmic Bias	Systematic unfairness in AI outputs affecting certain groups
Explainability (XAI)	The ability to understand and explain why an AI produced a specific output
Governance	Policies, laws, and oversight structures controlling AI development and deployment
Transparency	Openness about how AI systems work and make decisions
Accountability	The responsibility individuals and organisations bear for AI outcomes

Privacy	The right to control personal information and how it is collected and used
Responsible AI	AI developed and deployed according to ethical and safety standards
Black Box	An AI system whose internal reasoning cannot be inspected or explained
Risk-Based Governance	Regulating AI according to the severity of potential harms

WEEK 13 — AI INNOVATION FOR ZAMBIA & AFRICA, ENTREPRENEURSHIP & LOCAL AI SOLUTIONS

Theme: Using Generative AI to Solve Local Problems and Build African Innovation

1. Shifting the Mindset — From Consumer to Creator

Throughout the global AI story, Africa has largely been a **consumer** of technology built elsewhere. This week challenges you to imagine and begin building a different future — one where Zambia and Africa are **active creators and innovators** in AI, not just end-users of foreign tools.

The goal: Move from 'using foreign AI tools' toward 'building locally relevant AI solutions' that address African challenges with African context, languages, and values.

2. AI Opportunities in Zambia and Africa

Here are some of the most promising sectors for AI innovation in Zambia and Africa:

Agriculture

Zambia's economy depends heavily on farming. AI can provide crop disease detection (analyse photos of sick plants), weather prediction and advisory services, market price information to help smallholder farmers get fair prices, and soil analysis guidance.

Healthcare

AI can assist with medical diagnosis (especially in areas lacking specialists), health information chatbots in local languages, disease surveillance and prediction, and patient record management.

Education

AI tutoring assistants in local languages, personalised learning pathways, automated question generation for revision, and tools to support teachers in large classrooms.

Finance

AI for mobile money fraud detection, credit scoring for people without formal banking history, financial literacy chatbots, and insurance claim processing.

Governance & Public Services

AI-powered citizen information systems, document processing, infrastructure monitoring, and corruption detection.

Tourism

AI-powered tour guides, language translation, personalised itinerary planning, and cultural heritage preservation.

3. Local Language Technologies — A Critical Gap

One of the most important areas for African AI innovation is **natural language processing in local languages**. Most AI systems are trained predominantly on English, European, and East Asian languages. Languages like **Bemba, Nyanja, Tonga, and Lozi** are severely underrepresented in AI training data.

The consequences are real: a Zambian farmer asking for agricultural advice in Bemba gets a poor response or no response at all. A health information chatbot that only works in English excludes most rural users. Building AI that works in local languages is both a technical challenge and a matter of **digital justice and inclusion**.

4. Infrastructure Challenges — Be Realistic

AI innovation in Africa faces real infrastructure barriers that must be honestly acknowledged:

- **Connectivity** — many rural areas have limited or expensive internet access. AI applications that require constant cloud connectivity will not reach most users.
- **Compute Limitations** — training large AI models requires enormous computational resources that most African institutions cannot afford.
- **Electricity reliability** — data centres and AI systems require consistent power supply.
- **Cost Barriers** — commercial AI API calls cost money. At scale, this can be prohibitive for low-income contexts.
- **Data Scarcity** — there is relatively little labelled training data in African languages and contexts.

Design implication: The best AI solutions for Africa will often be lightweight, offline-capable, low-bandwidth, and built with open-source tools to keep costs down.

5. AI Entrepreneurship — Building AI Businesses

Generative AI has dramatically lowered the barrier to starting an AI-powered business. You no longer need to train a model from scratch — you can build valuable products by creatively combining existing APIs and tools. Key AI business models include:

- **AI-as-a-Service (AaaS)** — provide AI capabilities to other businesses through an API.
- **AI-enhanced products** — add AI features (smart search, summarisation, recommendations) to an existing product.
- **AI consulting and implementation** — help organisations deploy AI responsibly.
- **AI content and media** — use AI tools to produce content, designs, or media at scale.
- **AI for social impact** — build solutions for underserved communities and seek NGO or government funding.

6. AI Sovereignty

AI Sovereignty refers to a country's or region's ability to own, control, and govern its own AI capabilities and infrastructure. Without AI sovereignty, nations depend on foreign companies for critical digital infrastructure — creating risks of data exfiltration, service disruption, and loss of cultural and linguistic representation in AI systems.

7. Digital Inclusion and Responsible Innovation

- **Affordability** — solutions must be accessible to low-income users.
- **Accessibility** — design for users with limited literacy, different languages, or disabilities.
- **Community impact** — consider who benefits AND who may be excluded or harmed.
- **Sustainability** — AI systems require ongoing maintenance and energy; consider long-term costs.
- **Ethical entrepreneurship** — profit motives and social good must be balanced.

KEY TERM	MEANING
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AI for Development	Using AI technologies to support social and economic progress in developing contexts
AI Entrepreneurship	Building AI-driven businesses and products
AI Sovereignty	Local ownership and control of AI infrastructure and systems
Digital Inclusion	Ensuring broad access to digital technology across all populations
Localisation	Adapting AI systems to work effectively in local languages and cultural contexts
Innovation Ecosystem	The environment of companies, investors, universities, and policies that supports entrepreneurship
Sustainable AI	AI developed with long-term social, environmental, and economic sustainability in mind
Edge AI	Running lightweight AI on local devices without requiring constant internet
Open-Source AI	AI models and tools freely available for anyone to use, modify, and build on
Dataset Scarcity	The limited availability of quality training data for underrepresented languages and contexts

WEEK 14 — FINAL PROJECT DEMONSTRATIONS, VIVA PRESENTATIONS & COURSE REVISION

Theme: Demonstrating Competence, Creativity, Critical Thinking, and Responsible AI Innovation

1. What Week 14 Is About

Week 14 is the culmination of your entire semester. This is not just an ending — it is a **celebration and demonstration** of everything you have learned and built. You will showcase your final projects, defend your work in a viva, reflect on your growth, and contribute to peer learning.

Remember: You are expected to demonstrate not only WHAT you built, but WHY you built it, HOW it works, its limitations, and its societal implications.

2. Final Week Assessment Components

Component	Weighting	What You Do
Final Project Demonstration	40%	Live demo of your AI project — show it works, explain it, and discuss reliability
Viva / Oral Defence	20%	5-10 min individual questioning — defend your technical and ethical decisions
Final Reflection Portfolio	20%	Written reflection on skills learned, challenges faced, ethical insights, and growth
Participation, Peer Review & Engagement	20%	Actively engage in the expo, provide peer feedback, and contribute to discussions

3. What Makes a Strong Final Project

Your project should demonstrate all the skills you have built across the semester. Strong projects typically include all of these elements:

- Clear Problem Statement — what real challenge does your system address? Why does it matter?
- Thoughtful AI Workflow — prompts, retrieval, APIs, and multimodal components working together.
- Technical Demonstration — the system actually works and you can show it live.
- Reliability Discussion — honest assessment of hallucination risks and mitigation strategies.
- Ethical Analysis — bias, privacy, accessibility, and responsible usage are considered.
- Future Improvements — what would you do with more time? What are the current limitations?

4. Final Project Rubric

Criterion	Marks
Technical Functionality	20
Innovation and Creativity	15
Workflow Design	15
Prompt Engineering Quality	10
Reliability and Verification	10
Ethical Awareness	10
Presentation Quality	10

5. Viva Questions — Prepare Yourself

Practice answering these categories of questions before your viva:

Technical Questions:

- Why did you choose this architecture for your system?
- How does your retrieval system work?
- What causes hallucinations in your system and how did you mitigate them?
- How did prompt engineering affect your system's performance?
- Why did you select this particular AI model or API?

Ethical Questions:

- What bias risks exist in your system?
- How does your system protect user privacy?
- Could your system be misused — and what safeguards did you implement?
- Who might be excluded from or disadvantaged by your system?

Reflection Questions:

- What was your biggest technical or design challenge?
- What would you improve if you had more time?
- What surprised you most about working with AI systems?

6. The Reflection Portfolio

Your reflection portfolio is your opportunity to show intellectual growth throughout the semester. Cover all of these areas:

- Technical Skills Developed — prompt engineering, API usage, workflow design, RAG, multimodal AI.
- Critical Thinking Growth — how has your scepticism about AI outputs developed?
- Challenges Experienced — be honest about what was hard: debugging, unreliable outputs, prompt failures.
- Ethical Reflections — how should AI be developed and governed responsibly?
- Future Career Interests — what AI-related paths interest you? What do you want to learn next?

7. Course-Wide Summary — What You Should Know by Now

By the end of Week 14, you should have a strong understanding of all these topics:

- Generative AI fundamentals — what it is, how it works, and what it can and cannot do.
- Neural networks and deep learning — the technical foundation of modern AI.
- Transformers and attention mechanisms — the architecture behind LLMs.
- Large Language Models — how they are trained, deployed, and constrained.
- Prompt engineering — designing effective, reliable, and ethical prompts.
- AI hallucinations and evaluation — detecting and mitigating unreliable outputs.
- AI APIs and application development — building real AI-powered systems.
- Retrieval-Augmented Generation (RAG) — improving reliability through retrieval.

- Multimodal AI — text, image, speech, and video generation and analysis.
- AI ethics and governance — responsible development, fairness, and regulation.
- AI innovation for Zambia and Africa — local solutions, entrepreneurship, and AI sovereignty.

Final thought from your course: AI is a powerful tool, but human judgment, ethics, creativity, and critical thinking remain absolutely essential. Technology without wisdom is dangerous. You leave this course not just as AI users, but as critical, responsible AI practitioners.